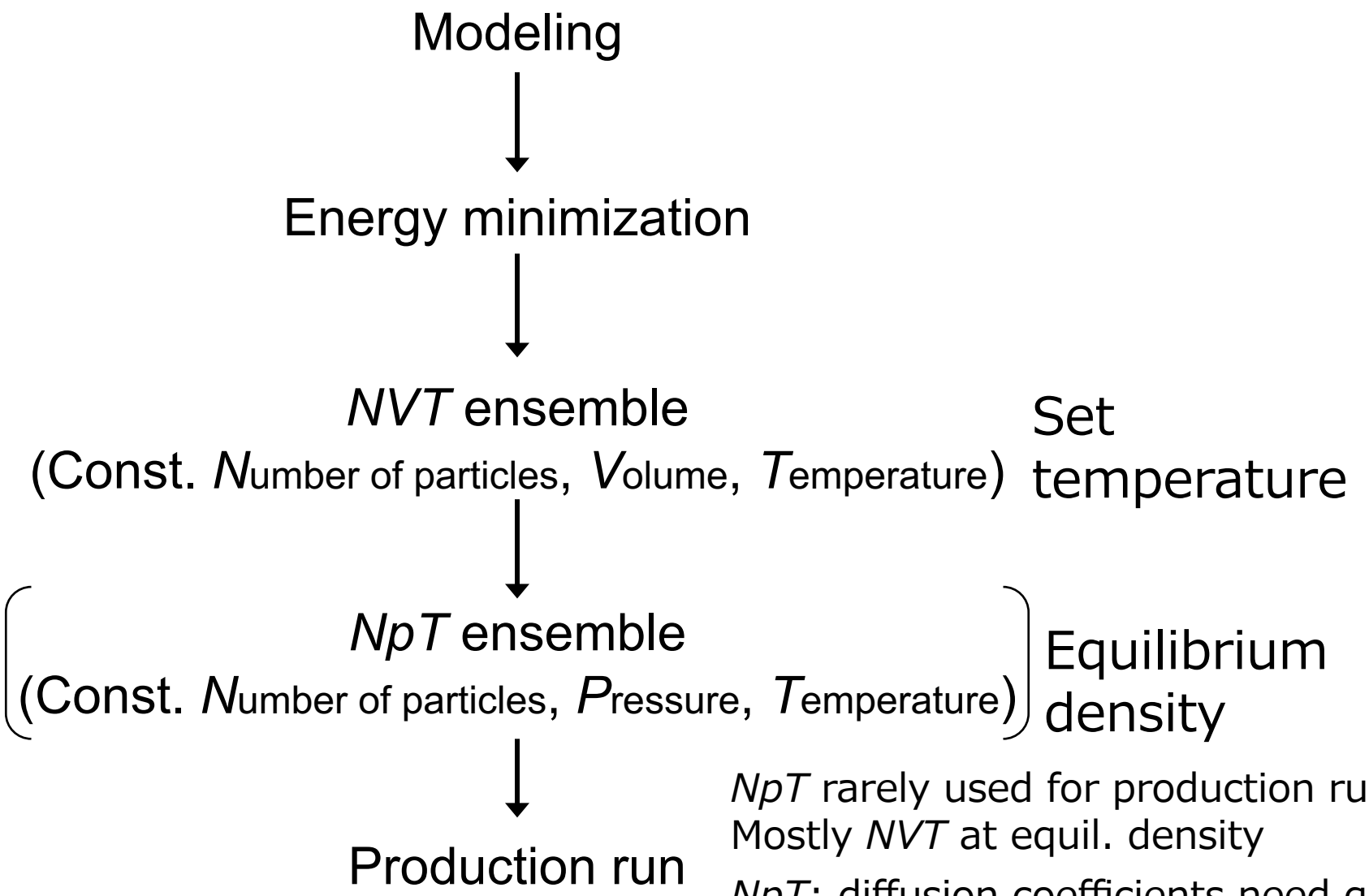
Introduction to Molecular Dynamics

Laboratory for Chemistry and Life Science, Institute of Integrated Research, Institute of Science Tokyo
Tateyama–Ando Lab. Ryoma Sasaki

1. Flow of MD simulation
2. Energy calculation
3. Time integration
4. Statistical mechanics; T & P control
5. Data analysis
   - **Static properties**
   - Dynamic properties

Initialization & Equilibration



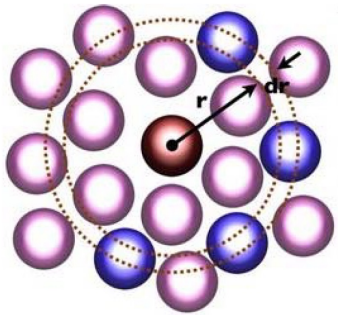
NpT rarely used for production run
Mostly *NVT* at equil. density

NpT: diffusion coefficients need great care
Hummer *et al.*, *JCTC*, **19**, 3406 (2023).

Radial distribution function

Radial distribution function (RDF; g(r): g-of-r)

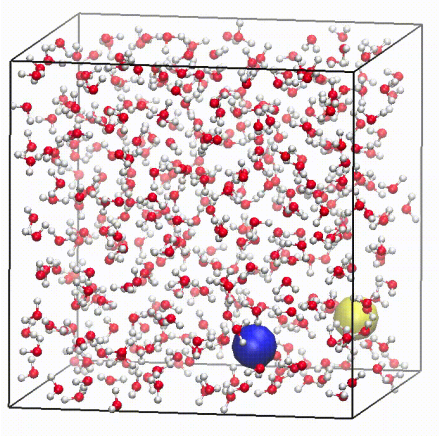
- How many other atoms are found around a given atom
- Also from X-ray/neutron data (pair distribution function)

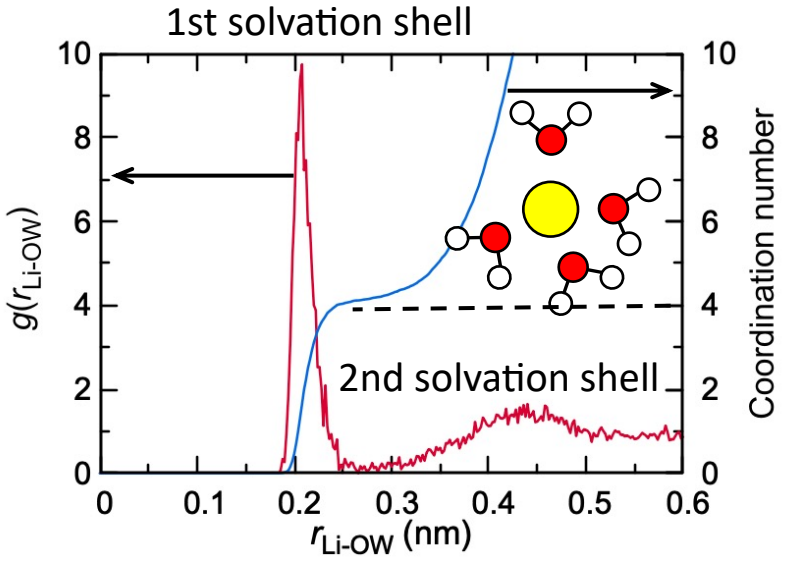


$$g(r) = \frac{\langle n(r) \rangle}{\rho \times 4\pi r^2 \mathrm{d}r} \qquad \rho = \mathrm{N/V} \qquad \mathrm{CN}(r) = \int_0^r g(r')\mathrm{d}r'$$

1. $r \to 0$ repulsion excludes particles: $g(r) \to 0$
2. $r \to \infty$ no positional correlation, mean density: $g(r) \to 1$

$1Li^+ + 1Cl^- + 400H_2O$ @300K
(2.3 nm x 2.3 nm x 2.3 nm)

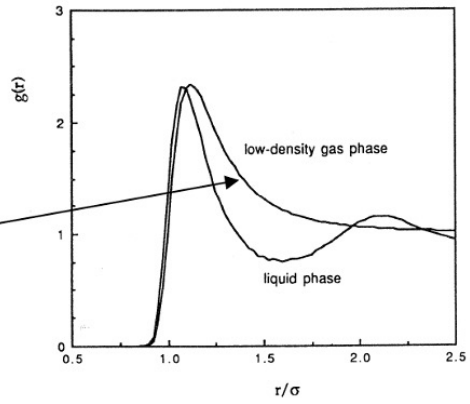




Radial distribution function

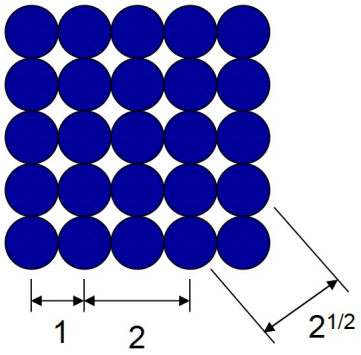
Low-density limit

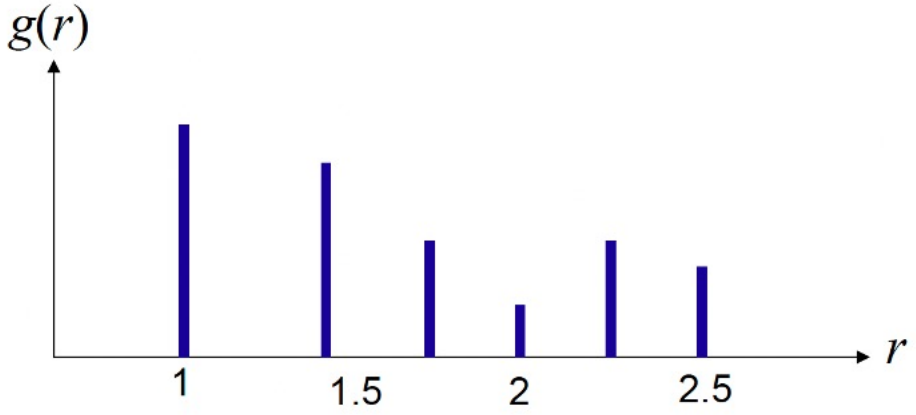
$$\lim_{\rho \to 0} g(r) = \exp\left[-\frac{U(r)}{kT}\right]$$



Crystal structure

Discrete values at lattice sites





Constant-volume heat capacity

$$C_v = \left(\frac{\partial \mathcal{H}}{\partial T}\right)_V$$

Method ①
- $\langle H(T) \rangle$ from *NVT* at several T
- $\langle H(T) \rangle$ vs. *T*: take the slope

Method ②
- From *U* fluctuation

$$C_v = \frac{\langle (\delta \mathcal{H})^2 \rangle}{k_\mathrm{B} T^2}$$

Def. of fluctuation

$$\delta \mathcal{H} := \mathcal{H} - \langle \mathcal{H} \rangle$$

Evaluating fluctuation magnitude requires long simulations
⇨ Method ① is generally more accurate

When T < Debye temp. (Einstein temp.), vibrational heat capacity needs a quantum correction*

*For solids, solid-state physics makes this simple, but for liquids, methods are still developing *JCTC*, **17**, 7187 (2021).

Also obtained like the heat capacity:

- Adiabatic compressibility $\kappa = \frac{-1}{V}\left(\frac{\partial V}{\partial P}\right)_{S=const}$
- Thermal pressure coefficient $\gamma = \left(\frac{\partial P}{\partial T}\right)_{V=const}$
- Etc.

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   - **Dynamic properties**

Dynamic properties

- Time correlation functions
  - Velocity autocorrelation function
  - Wiener–Khinchin theorem (vibrational density of states)

- Transport coefficients
  - Einstein relation (self-diffusion coefficient)
  - Green-Kubo formula
    - Self-diffusion coefficient
    - Ionic conductivity
  - Non-equilibrium MD

Time correlation function

Time correlation function of *A*, *B*

How quantity $A(t_0)$ relates to the value t later, $B(t_0 + t)$

$$C(t) = \lim_{\tau\to\infty} \frac{1}{\tau} \int_0^{\tau} A(t_0)B(t_0+t)dt_0 = \left\langle A(t_0)B(t_0+t)\right\rangle$$

($\frac{1}{\tau}\int_0^{\tau} \cdots \mathrm{d}t_0$: average over snapshots, shifting the time origin t_0)

✓ Invariant under shifts of the time origin, so

$$C(t) = \langle A(t_0)B(t_0+t)\rangle = \langle A(0)B(t)\rangle$$

- $t \to 0$ $\quad C(0) = \lim_{\tau\to\infty} \frac{1}{\tau}\int_0^{\tau} A(t_0)\,B(t_0)\,dt_0 = \langle A(t_0)\,B(t_0)\rangle \equiv \langle AB\rangle$

- $t \to \infty$ (Unless A,B vary periodically) A,B become uncorrelated, so

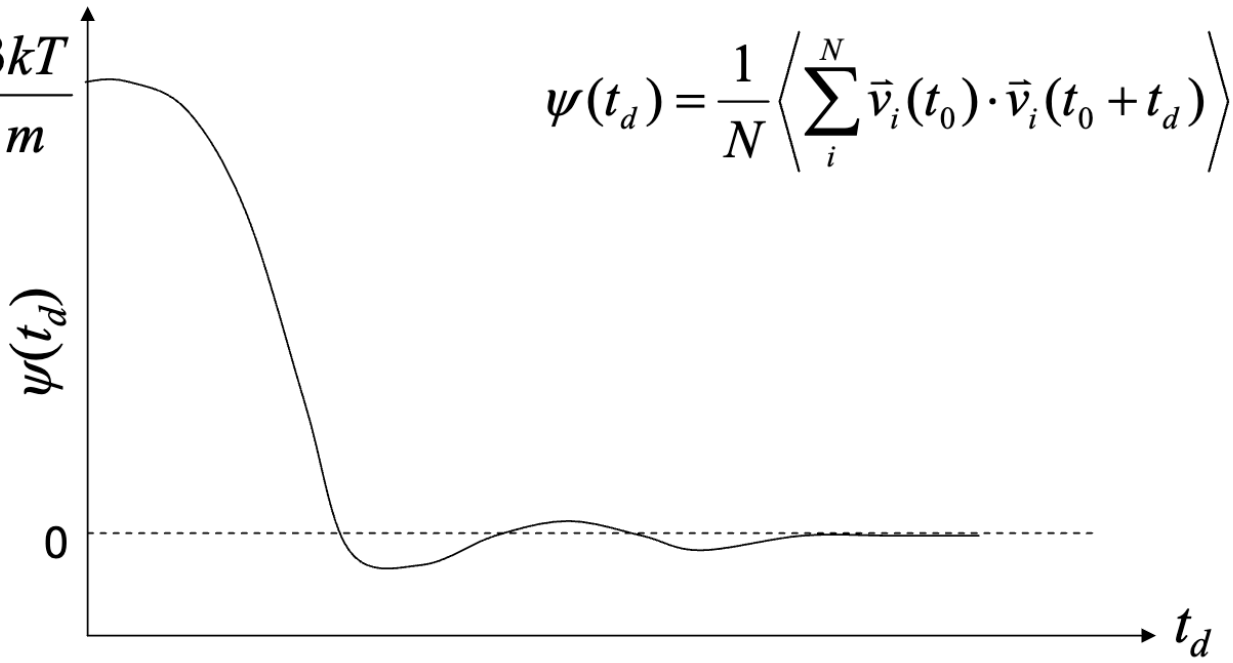
$$\lim_{t\,\to\,\text{large}} C(t) = \langle A\rangle\langle B\rangle$$

Velocity autocorrelation function

- Single particle correlations
  - Dynamic quantity is a property of individual particles
- Velocity autocorrelation function

$$\psi(0) = \frac{3kT}{m}$$

$$\psi(t_d) = \frac{1}{N}\left\langle \sum_i^N \vec{v}_i(t_0)\cdot\vec{v}_i(t_0+t_d)\right\rangle$$



Gives a rough relaxation time, but little other information
→ Fourier transform → discuss in frequency (wavenumber) domain

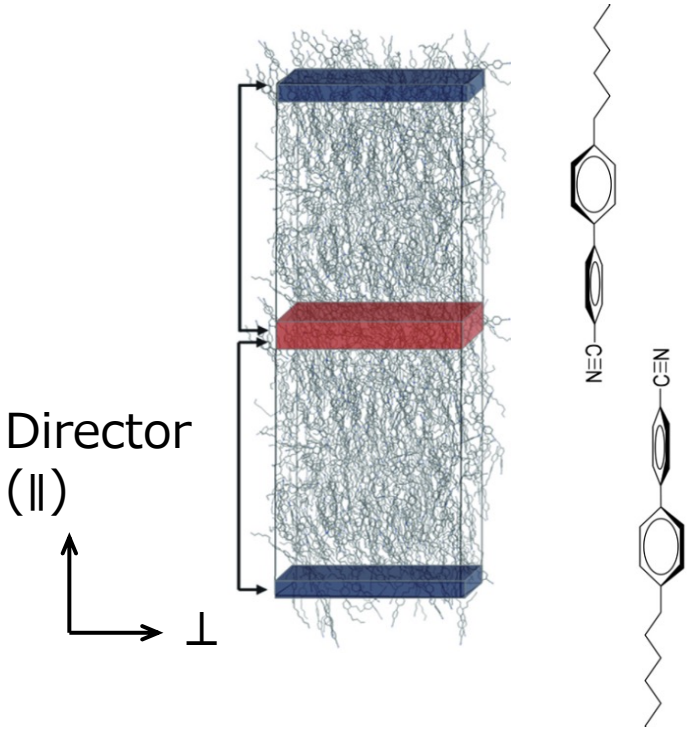
Wiener–Khinchin theorem (vibrational density of states)

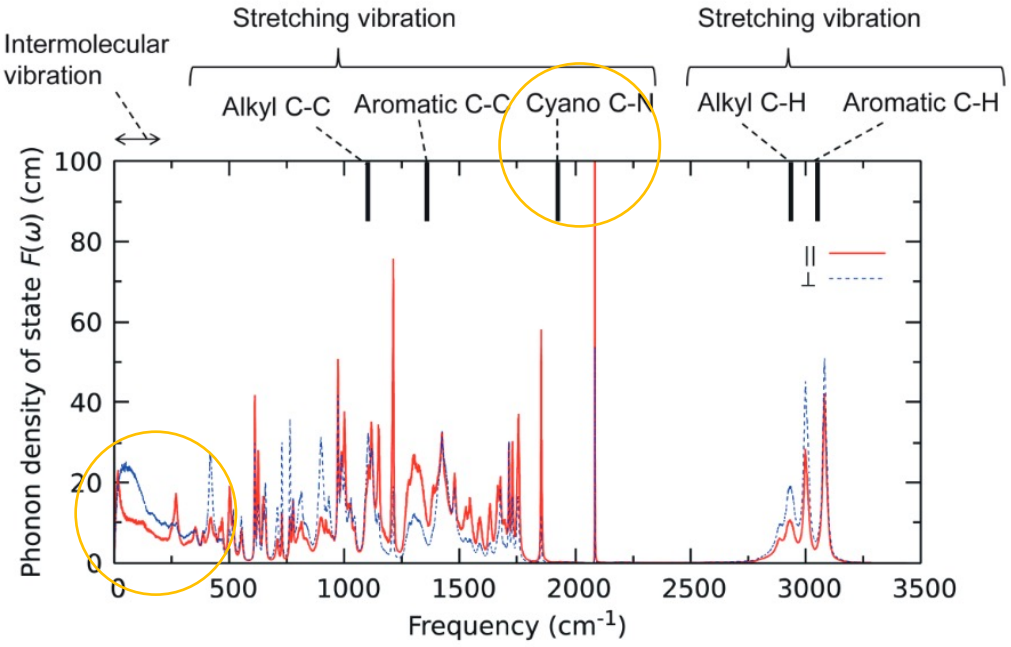
$$I(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \langle A(t)A(0) \rangle \exp(i\omega t) \mathrm{d}t$$

Fourier-transforming the velocity autocorrelation function gives the vibrational density of states (VDOS)

~Choosing time step Δt ~IR, Raman, etc…

Below: vibrational anisotropy in a liquid-crystal phase





Dynamic properties

- Time correlation functions
  - Velocity autocorrelation function
  - Wiener–Khinchin theorem (vibrational density of states)

- Transport coefficients
  - Einstein relation (self-diffusion coefficient)
  - Green-Kubo formula
    - Self-diffusion coefficient
    - Ionic conductivity
  - Non-equilibrium MD

Transport coefficients

Transport coefficients…

$$\text{Flux} = -(\text{transport coefficient}) \times \text{gradient}$$

Current	Elec. conductivity	Electric field ($-\nabla\varphi$)	Ohm's law
Heat flux	Thermal conductivity	Temperature gradient	Fourier's law
Shear stress	Viscosity	Velocity gradient	Newton's law of viscosity
Mass flux	Diffusion coefficient (strictly, chemical diffusivity)	Concentration gradient	Fick's law

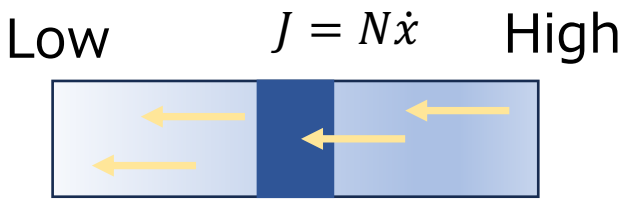
Calculation methods

- Einstein relation
- Green-Kubo formula
- Non-equilibrium MD (NEMD)

Self-diffusion coefficient

– Fick's law and conservation of mass

$$N\dot{x} = -D\frac{\partial N}{\partial x} \qquad \frac{\partial N}{\partial t} + \frac{\partial (N\dot{x})}{\partial x} = 0$$



– Combine and solve for N

$$\frac{\partial N}{\partial t} = D\frac{\partial^2 N}{\partial x^2} \qquad N(x,t) = \frac{N_0}{2\sqrt{\pi D t}}\exp\left[\frac{-x^2}{4Dt}\right]$$

– Second moment of the distribution is the mean-square displacement

$$\left\langle [x(t) - x(0)]^2 \right\rangle = \frac{1}{N_0}\int x^2 N(x,t)dx = 2Dt$$

$t \to \infty$ limit:

$$D = \lim_{t\to\infty} \frac{\left\langle [x(t) - x(0)]^2 \right\rangle}{2t}$$

(Assumption: diffusion coefficient= const over this concentration range)
i.e., ~ dilute system – no concentration effects: solutes don't interact
c.f. concentrated system – solute–solute interactions not negligible

“Calculating” the self-diffusion coefficient

◎ Dimensionality

1D

$$D = \lim_{t\to\infty} \frac{\left\langle [x(t) - x(0)]^2 \right\rangle}{2t}$$

(d-dim. $D = \lim_{t\to\infty} \frac{\langle [\boldsymbol{r}(t) - \boldsymbol{r}(0)]^2 \rangle}{2dt}$)

3D

$$D = \frac{1}{3}\left[D_x + D_y + D_z\right] = \lim_{t\to\infty} \frac{\langle [\boldsymbol{r}(t) - \boldsymbol{r}(0)]^2 \rangle}{3\times 2t}$$

$$= \lim_{t\to\infty} \frac{\langle [\boldsymbol{r}(t) - \boldsymbol{r}(0)]^2 \rangle}{6t}$$

✓ Averaging over 3 directions improves statistical accuracy

◎ MSD calculation

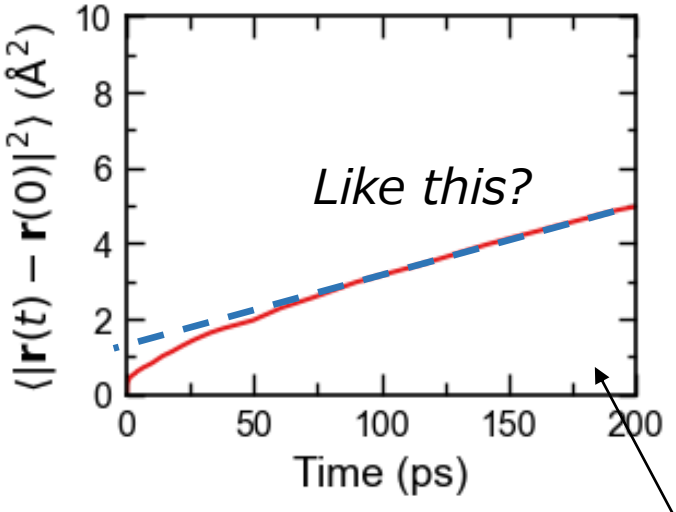
$$\langle [\boldsymbol{r}(t) - \boldsymbol{r}(0)]^2 \rangle = \frac{1}{N}\sum_i \langle [\boldsymbol{r}_i(t) - \boldsymbol{r}_i(0)]^2 \rangle = \frac{1}{N}\sum_i \frac{1}{\tau}\int_0^\tau \{r_i(t_0 + \mathrm{t}) - r_i(t_0)\}^2 \mathrm{d}t_0$$

Average over same-species particles for better statistics

Shift the time origin c.f. time correlation function

“Calculating” the self-diffusion coefficient

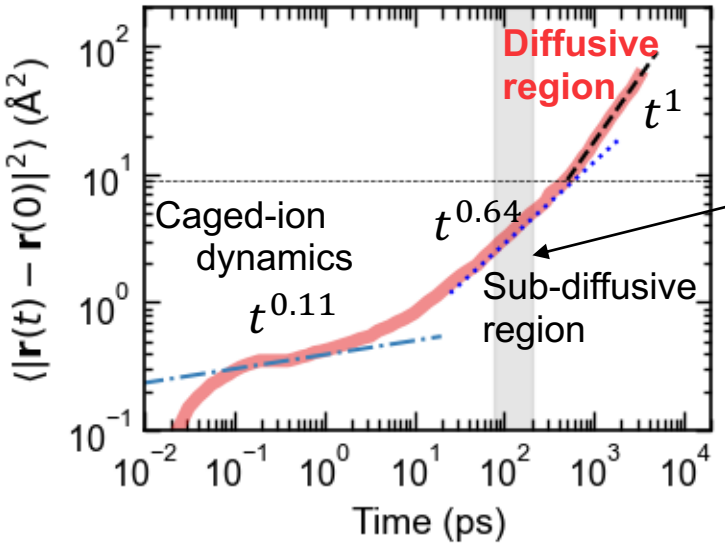
Q. How long is t→∞ in practice? (e.g., Li solid electrolyte)



$$D = \lim_{t\to\infty} \frac{\langle[\boldsymbol{r}(t)-\boldsymbol{r}(0)]^2\rangle}{6t}$$

$$\langle[\boldsymbol{r}(t)-\boldsymbol{r}(0)]^2\rangle = 6Dt \qquad (t\to\infty)$$

D = const. → MSD linear in t



Keyword: Diffusive region

$$\log\langle[\boldsymbol{r}(t)-\boldsymbol{r}(0)]^2\rangle = \log t + b \qquad \text{MSD} \propto t^1$$

log-log plot shows it is not $\propto t^1$

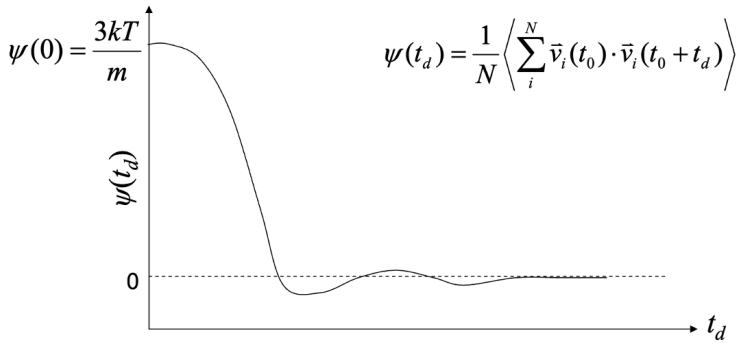
In the region above, MSD is simply too small
Li-Li distance in solid ~ 3Å → MSD ≈ (3Å)^2

Roughly once MSD reaches this distance:
”Diffusive region”

Green-Kubo formula

Flux J = − (transport coeff.) x gradient

$$(\text{coeff.}) \propto \int_0^\infty \langle J(t)J(0)\rangle \, \mathrm{d}t$$



Self-diffusion coefficient

$$D = \frac{1}{d}\int_0^\infty \langle J(t)J(0)\rangle \, \mathrm{d}t \qquad J(t) = v(t)$$

Ionic conductivity

$$\sigma = \frac{e^2}{dVk_\mathrm{B}T}\int_0^\infty \langle J_\mathrm{q}(t)J_\mathrm{q}(0)\rangle \, \mathrm{d}t \qquad J_\mathrm{q}(t) = \sum_i q_i v_i(t)$$

q_i: formal charge

Transport coefficients from equilibrium fluctuations

However, D, σ are usually computed with MSD-based methods

* *Time correlation functions are large only at short times*

→ • Trajectory storage becomes large • Analysis time

• *Hard to judge convergence of the integral etc…*

Self-diffusion coefficient and ionic conductivity

Self-diffusion coefficient $D = \frac{1}{d}\int_0^\infty \langle J(t)J(0)\rangle \, \mathrm{d}t$ $J(t) = v(t)$

Ionic conductivity $\sigma = \int_0^\infty \langle J_\mathrm{q}(t)J_\mathrm{q}(0)\rangle \, \mathrm{d}t$ $J_\mathrm{q}(t) = \sum_i q_i v_i(t)$

$$\sigma \propto \int_0^\infty \left[\sum_i q_i^2 \langle v_i(t)v_i(0)\rangle + \sum_{i,j\neq i} q_i q_j \langle v_i(t)v_j(0)\rangle \right] \mathrm{d}t$$

	Single-particle picture	Ion–ion correlation	
Dilute electrolyte	✓	= 0	
Conc. electrolyte **Solid electrolyte**	✓	✓	Important in dense materials
PFG-NMR	✓	–	
Impedance spectrum	✓	✓	
	∝ self-diff. coeff. *D*	Cost: high; long sampling required	

$\sum_{i,j\neq i} q_i q_j \langle v_i(t)v_j(0)\rangle$ is costly, so it is often approximated as 0

Nernst–Einstein approximation

Supplement: non-equilibrium MD (NEMD)

Apply field X explicitly; from the driven flux J, compute L directly

$$J = LX$$

J: heat flux L: thermal conductivity X: temp. gradient
J: ionic flux L: ionic conductivity X: electric field

[Pros]
- Usually more efficient than Green-Kubo: $\mathrm{Var}(L) \propto X^2 t$ Stoltz *et al.*, *Multiscale Model. Simul.*, **10**, 191 (2012)
- Can also capture nonlinear response

$J = L_1 X + L_2 X^2 + \cdots$

[Cons]
- Anisotropic systems need multiple runs
- Usual transport coefficients: linear-response regime

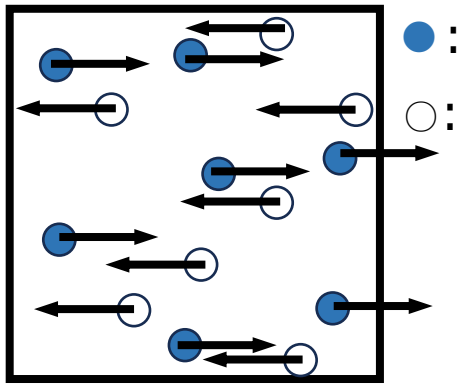
  → Verify linear response, i.e., $X \to 0$: $J = L_1 X + \mathcal{O}(X^2)$ *with* $\mathcal{O}(X^2) \sim 0$ *for the applied field (extra effort)*

Supplement: NEMD for self-diffusion coefficient

Color-diffusion NEMD

Fictitious / Color charge $c_i = (-1)^i$

$$\begin{cases} \dfrac{\mathrm{d}\boldsymbol{r}_i}{\mathrm{d}t} = \dfrac{\boldsymbol{p}_i}{m_i} \\ \dfrac{\mathrm{d}\boldsymbol{p}_i}{\mathrm{d}t} = \boldsymbol{F}_i + c_i \boldsymbol{F}_\mathrm{e} + \alpha \boldsymbol{p}_i \end{cases}$$



Flux $$J_\mathrm{c}(t) = \sum_{i \in \mathrm{Color}} (-1)^i v_i(t)$$

* Momentum conservation:
N(c_i=1) = N(c_i =−1)

Self-diffusion coefficient $$D_\mathrm{self} = \frac{k_\mathrm{B}T}{N/V} \lim_{F_\mathrm{e}\to 0, t\to\infty} \frac{\langle J_\mathrm{c}(t)\rangle_{F_\mathrm{e}}}{F_\mathrm{e}}$$

(Thermodynamic limit: N/V = const., V→∞)

Evans *et al.*, Phys. Rev. A, 28, 1016 (1983).
Evans *et al.*, Statistical Mechanics of Nonequilibrium Liquids (Cambridge University Press).
Searles *et al.*, *J. Chem. Theory Comput.*, **18**, 6, 3357–3363 (2022).

Supplement: NEMD for thermal conductivity

Reverse NEMD

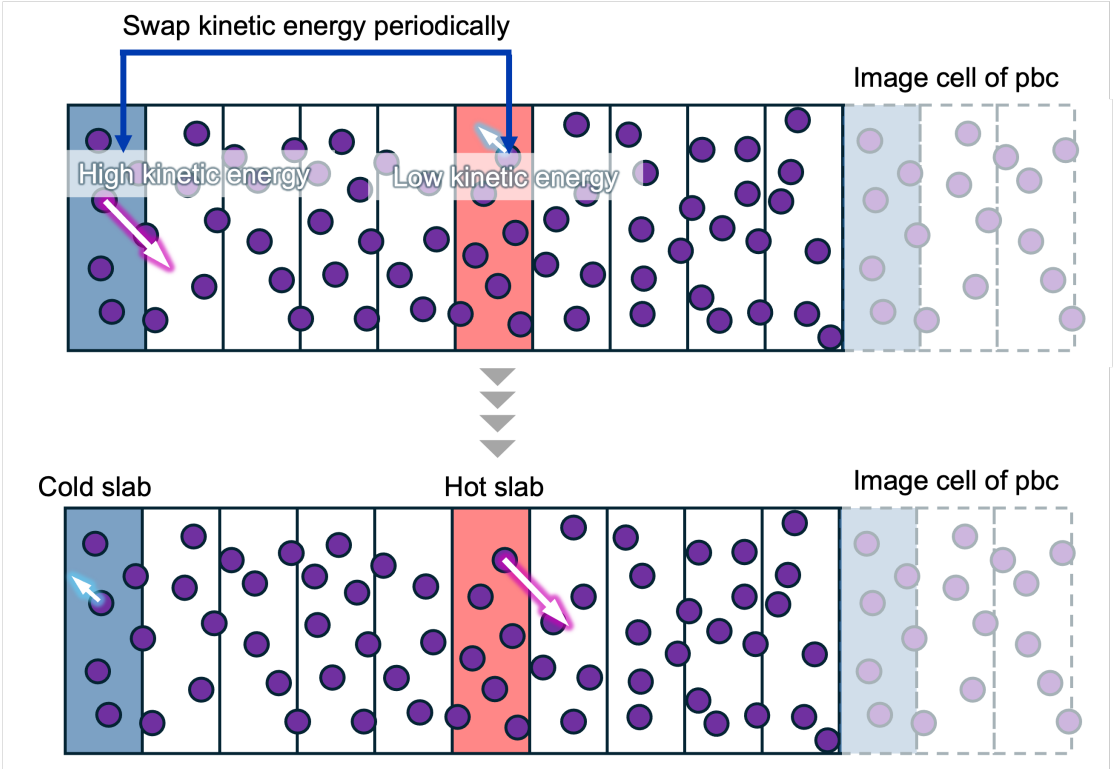
Microscopic heat flux

$$\mathbf{J}_e = \sum_I \epsilon_I \mathbf{V}_I$$ ***Often hard to derive***

$$-\frac{1}{2}\sum_{I,J\neq I}\left(\mathbf{V}_I \cdot \nabla_I v(|\mathbf{R}_I - \mathbf{R}_J|)\right)(\mathbf{R}_I - \mathbf{R}_J)$$

For atom I:
ε_I: energy, V_I: velocity, R_I: position
v: interatomic potential

Baroni *et al.*, *J. Low. Temp. Phys.*, **185**, 79 (2016).



Heat flux via kinetic-energy swaps

Heat flux $$J_q = \frac{1}{2tS}\sum_{\text{transfer}} \frac{m}{2}\left(v_h^2 - v_c^2\right)$$

Fourier's law $$\kappa = -\frac{J_q}{\langle \partial T/\partial z \rangle}$$

Müller-Plathe, *J. Chem. Phys.*, **106**, 6082 (1997).

Assignment

Q1 or Q2: answer either one (both is also OK)

Q1. Derive $D_{\mathrm{self}} = \int_0^\infty \langle v(t)v(0)\rangle \mathrm{d}t = \lim_{t\to\infty} \frac{1}{2t}\langle |x(t) - x(0)|^2\rangle$.

Q2. Derive $C_v = \frac{\langle (\delta\mathcal{H})^2\rangle}{k_\mathrm{B}T^2}$ for the heat capacity.

- By def. $C_v = \left(\frac{\partial \mathcal{H}}{\partial T}\right)_V$ so $C_v = \frac{\partial \langle H\rangle_{NVT}}{\partial T}$: carry this out to derive it
- *NVT* distribution function: $f_{NVT}(\mathbf{r},\mathbf{p}) = \frac{\exp[-\mathcal{H}(\mathbf{r},\mathbf{p})/(k_\mathrm{B}T)]}{\iint \mathrm{d}\mathbf{r}\mathrm{d}\mathbf{p}\exp[-\mathcal{H}((\mathbf{r},\mathbf{p}))/(k_\mathrm{B}T)]}$.

$$\langle H\rangle_{NVT} = \int f_{NVT}(\mathbf{r},\mathbf{p}) H(\mathbf{r},\mathbf{p}) \mathrm{d}\mathbf{r}\mathrm{d}\mathbf{p}$$